Week 4 – Order Management System

1. Orders & Order_Details Table Design

Orders Table

The Orders table stores information about customer orders.

Column Name	Data Type	Constraint
Order_ID	INT	Primary Key
Customer_ID	INT	Foreign Key, NOT NULL
Order_Date	DATE	NOT NULL
Total_Amount	DECIMAL(10,2)	NOT NULL, CHECK
Order_Status	VARCHAR(20)	NOT NULL, CHECK

Order_Details Table

The Order_Details table stores the products included in each order.

Column Name	Data Type	Constraint
Order_Detail_ID	INT	Primary Key
Order_ID	INT	Foreign Key, NOT NULL
Product_ID	INT	Foreign Key, NOT NULL
Quantity	INT	NOT NULL, CHECK
Price	DECIMAL(10,2)	NOT NULL, CHECK

Constraints Used:

- Primary Key uniquely identifies each record.
- Foreign Key connects Orders with Customer and Order_Details, and Product with Order_Details.
- NOT NULL ensures required fields are not empty.
- UNIQUE prevents duplicate values or duplicate product entries within the same order.

- CHECK validates order status, quantity, price and total amount.

2. Updated ER Diagram

The Entity Relationship (ER) diagram represents the relationships between the four tables.

- Customer → Orders: One customer can place many orders.
- Orders → Order_Details: One order can contain multiple products.
- Product → Order_Details: One product can be included in many orders.

ER Diagram representation:

CUSTOMER

|

| 1 : M

|

ORDERS

|

| 1 : M

|

ORDER_DETAILS

|

| M : 1

|

PRODUCT

Primary keys and foreign keys maintain relationships and ensure data consistency across the tables.

3. Sample Customer, Order & Order_Details Data

Customer Table

Customer_ID	Customer_Name	Address
101	Arun Kumar	Chennai

102	Priya Devi	Coimbatore
103	Karthik Raj	Madurai
104	Meena S	Salem

Orders Table

Order_ID	Customer_ID	Order_Date	Total_Amount	Order_Status
301	101	2026-10-01	1550.00	Pending
302	102	2026-10-03	2300.00	Shipped
303	101	2026-10-10	2200.00	Delivered
304	103	2026-10-12	1500.00	Pending

Order_Details Table

Order_Detail_ID	Order_ID	Product_ID	Quantity	Price
401	301	201	1	750.00
402	301	202	2	400.00
403	302	203	1	1200.00
404	302	204	2	550.00
405	303	205	1	1800.00
406	303	202	1	400.00
407	304	201	2	750.00

Note: These are illustrative sample records. Order totals must match the sum of Quantity × Price for each order.

4. Customer Order History & Purchase Analysis Report

The Customer Order History report displays customer details, order dates, order status and total amounts. It helps track the purchasing activities of each customer.

The Product-wise Order Report identifies the products ordered by customers, their quantities and prices. It helps understand product demand and purchasing patterns.

The Number of Times Each Product Was Ordered report counts how frequently each product appears in orders.

The Total Quantity of Products Sold report calculates the quantity sold for each product, excluding cancelled orders.

The Maximum Orders by Customer report identifies customers who have placed the highest number of non-cancelled orders.

The Total Spending by Customer report calculates the amount spent by each customer. It helps identify customers with higher purchase totals.

The Average Order Value report calculates the average amount spent per non-cancelled order.

The Total Sales Amount report calculates the total value of all non-cancelled orders. It helps understand the overall sales performance of the business.

5. Query Output Screenshots

Screenshots of the SQL query results from MySQL Workbench are included as proof of successful execution.

The screenshots should cover:

1. Orders and Order_Details table structures.
2. Sample customer, order and order-detail records.
3. Complete order details with customer and product information.
4. Customer order history report.
5. Product-wise order report.
6. Number of times each product was ordered.
7. Total quantity of products sold.
8. Customers with the maximum number of orders.
9. Total spending by each customer.
10. Average order value and total sales amount.

These outputs demonstrate that the Order Management System successfully stores order information, maintains relationships, manages order operations and generates useful purchase analysis reports.